

Gummi 0.8.3

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Asynchronous Bayesian Multi-Sensor EKF with Moment Matching

1. State Space Model

Let the latent state be

$$x_k \sim \mathcal{N}(\hat{x}_{k|k}, P_{k|k})$$

Dynamics:

$$x_k = f(x_{k-1}, u_{k-1}) + w_{k-1}, \quad w \sim \mathcal{N}(0, Q)$$

Sensor i :

$$z_k^{(i)} = h_i(x_k) + v_k^{(i)}, \quad v^{(i)} \sim \mathcal{N}(0, R^{(i)})$$

2. EKF Prediction (global clock)

For arbitrary time Δt :

$$\hat{x}_{k|k-1} = f(\hat{x}_{k-1|k-1}, u_{k-1})$$

$$P_{k|k-1} = F_k P_{k-1|k-1} F_k^T + Q_k$$

3. Asynchronous Updates

Each sensor has its own timestamps $t_j^{(i)}$.

When sensor i arrives:

Sequential Bayesian update:

$$p(x|z^{(1)}, z^{(2)}, \dots) \propto p(x) \prod_i p(z^{(i)}|x)$$

Implement sequentially:

$$(\hat{x}, P) \leftarrow \text{EKF_update}_i(\hat{x}, P, z^{(i)})$$

This is equivalent to a joint update under independence.

4. Local Sensor Filters (Optional)

Each sensor runs its own EKF:

$$x^{(i)} \sim \mathcal{N}(\mu_i, \Sigma_i)$$

This yields local posteriors:

$$p_i(x) = \mathcal{N}(\mu_i, \Sigma_i)$$

5. Bayesian Fusion via Moment Matching

Goal: combine $\{p_i(x)\}$ into global posterior.

Product of Gaussians:

$$\Sigma^{-1} = \sum_i \Sigma_i^{-1}$$

$$\mu = \Sigma \sum_i \Sigma_i^{-1} \mu_i$$

Moment matching (robust alternative):

$$\mu = \sum_i w_i \mu_i$$

$$\Sigma = \sum_i w_i (\Sigma_i + (\mu_i - \mu)(\mu_i - \mu)^T)$$

Weights:

$$w_i \propto \Sigma_i^{-1}$$

6. Prior Construction (Bayesian view)

Recursive prior:

$$p(x_k) = \int p(x_k | x_{k-1}) p(x_{k-1} | z_{1:k-1}) dx_{k-1}$$

Approximation:

$$p(x_k) \approx \mathcal{N}(\hat{x}_{k|k-1}, P_{k|k-1})$$

7. Handling Asynchrony

For sensor with delay τ :

Option A: propagate forward

$$x(t_k) = \Phi(\tau)x(t_k - \tau)$$

Option B: out-of-sequence update (OOSM)

Maintain buffer of past states and re-update.

Option C (simple): process measurements in arrival order with variable Δt .

8. Minimal Algorithm

Loop over time:

1. Predict with system model
2. For each arriving sensor measurement:
 - propagate to measurement time
 - EKF update (sequential)
3. Optional: fuse multiple sensor posteriors via moment matching

9. Remarks

- Sequential updates = exact for independent Gaussian likelihoods - Asynchronous sampling handled via variable-step prediction - Moment matching prevents overconfidence vs. naive fusion